of these technologies will result in intelligent products, smarter services and transformed business operations.

Qualities of a software architect

Software architects (SAs) lay the foundation for all architecture-related activities within the organisation. They work with the business and IT teams in an organisation to promote business agility, digital acceleration, reliable IT functions and better ROI across the organisation. They build solutions according to software principles and standards, and jointly develop reference artifacts for use across the organisation.

Successful SAs of the future must acquire a few important qualities to guide an organisation to success (Figure 1).

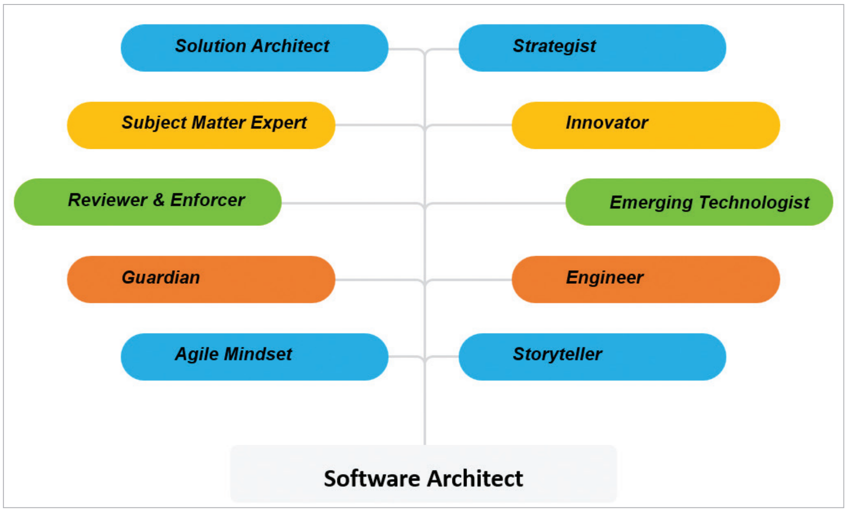

Figure 1: Role of a software architect

Strategist: As strategists, SAs work closely with the business and strategic governance teams in understanding the IT strategies, directions and initiatives. They provide recommendations to the strategic governance team on technology directions and analyse the impact of new initiatives on the enterprise architecture. They define a long-term technology roadmap by mastering new technology trends, and interacting with industry standards bodies and consortia to help shape standards and processes.

Innovator: As innovators, architects with high-level design skills move beyond imperative thinking to being disruptive innovators. They have the capability to be involved in product selection and integration in the broader landscape and guide projects across the organisation. By thoroughly researching the technology, and the resulting ideation, SAs provide internal teams and IT leaders with more than just a list of 'cool' technology ideas.

Emerging technologist: As emerging technologists, SAs have deep technology expertise and are aware of the latest developments and innovations. They solve business problems by investigating new and emerging technologies as well as innovations in pilot and proof-of-concept projects. They are tech savvy, aware, and can advise the chief digital officer (CDO) on the usage of APIs, microservices, cloud, DevOps, smart database technologies and other emerging trends. They also understand the importance of staying abreast of DevSecOps and emerging applications that could save the organisation time and money, and open up unlimited possibilities.

Engineer: As engineers, SAs bring efficiency, agility and automation to systems. Systems with efficiency and agility goals need the best engineering supported by the right architecture. Some of the key engineering skills that they need to master are:

- DevSecOps adoption, which helps drive agility and automation required for the business.
- Learn multiple programming languages and stacks to appreciate and take the best architecture decisions.
- Data engineering to understand data management and analytics for developing smart applications.
- Failure analysis to build resilient systems by understanding the request paths and failure modes.

Good communicator: Software architects communicate their strategy well and rally various groups across the organisation. They understand their audience and keep the solutions interesting and simple. Their communication addresses questions like why, who, what, when, where and how.

Agile architect: SAs promote the agile approach across the enterprise. As a product owner, the agile architect identifies the architecture required by an organisation. The focus is on architecture as a collaboration and building the simplest architecture that can possibly work. SAs must not under architect or over design, and isolate rapidly changing components from more stable ones. They build, test, implement architectural flow, and guide the team towards achieving quality deliverables on time. They stay invested with the team throughout, getting involved right from the point of functional specifications to make sure that what's expected is what's written. They also continuously check with the team to make sure they are not deviating from the stated design. Many a times, an architect may also have to address unwanted bureaucracy.

Guardian: As guardians, SAs ensure that standards are published, updated, and made readily available. They define architecture and technology standards for all